

# CHARCOAT

## PASSIVE FIRE PROTECTION

### CharCoat IC



### Optimum Thickness (DFT)

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## OPTIMUM THICKNESS FOR CHARCOAT IC INSULATOR COATING

### INTRODUCTION

The usage of silicone rubber coatings for high voltage insulators has increased exponentially in recent years and has replaced the employment of greases and, to a large extent, spray washing, as a solution to pollution flashover problems. In spite of its popularity, however, confusion regarding the optimum thickness of the coating layer is clearly evident.

Unfortunately, there is no international or national standard that addresses this issue and thus a wide variety of utility and manufacturer's specifications are in use. The only guidance from an independent organization is the IEEE Standard No. 1523. Unfortunately, on the subject of coating thickness, the content of this document is contradictory – some of the statements being:

*"The nominal thickness suggested by several manufacturers has been 0.5mm. This thickness...has been used for years in the coating industry and as such has little significance for the coating of insulators"*

*"The experience to date suggests that coating thickness is not a factor in either the performance or life of the coating"*

*"Coating thickness in the range of 0.125 to 0.7mm have been applied in the field with equal success"*

*"Laboratory tests indicate that when leakage currents of a damaging magnitude develop on coatings, then thickness plays a role in the coating life"*

*"Thick coatings...do not allow the heat to be conducted away to the porcelain substrate as quickly as thinner coatings"*

*"Thick coatings can result in higher hot spot temperatures during dry band arcing, thereby causing thermal degradation of the coating sooner than thinner coatings"*

*"A coating that is too thin, however, can also degrade quickly due to wearing from environmental forces"*

*"Tests indicate a thickness in the range of 0.38mm is optimum"*

It is clear from the above, and not surprising, that consensus on the best coating thickness to have has not been reached.

### COMMON COATING PRACTICE

Based on extensive testing at the coastal Koeberg Insulator Pollution Testing Station operated by South Africa's national utility, Eskom, and service experienced gained over a period of 26 years in all types of environment, we consider that coating thicknesses adopted in most cases are too great. There are many reasons why thick layers are applied, including:

- A general feeling that "thicker is better".
- Adoption of thicknesses traditionally specified for corrosion protection coatings
- Some coating types have a high viscosity owing to a large filler content and thus a thin layer is hard to achieve
- Manufacturers benefit by supplying larger volumes of material.

The effects of this are described below.

## FILLER CONTENT

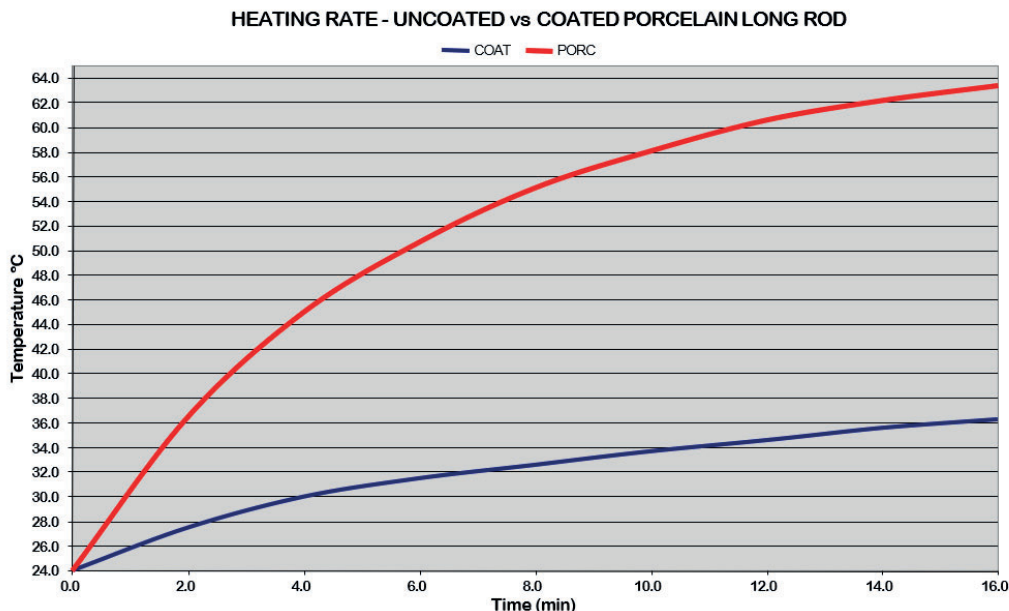
The amount of filler contained in the coating is a most important parameter. For good electrical performance and life, the correct balance between silicone polymer – which dictates the hydrophobicity, hydrophobicity transfer and hydrophobicity recovery rates – and filler – which protects the material from degradation caused by surface partial discharges – must be provided. There is, though, a tendency to include excess filler as it is significantly less expensive than silicone and improves the results of artificial laboratory tracking and erosion tests which intentionally destroy the hydrophobicity of the material under test. The thick layer naturally obtained with a coating containing a large amount of filler will not perform as well as a thin layer with a higher proportion of silicone.

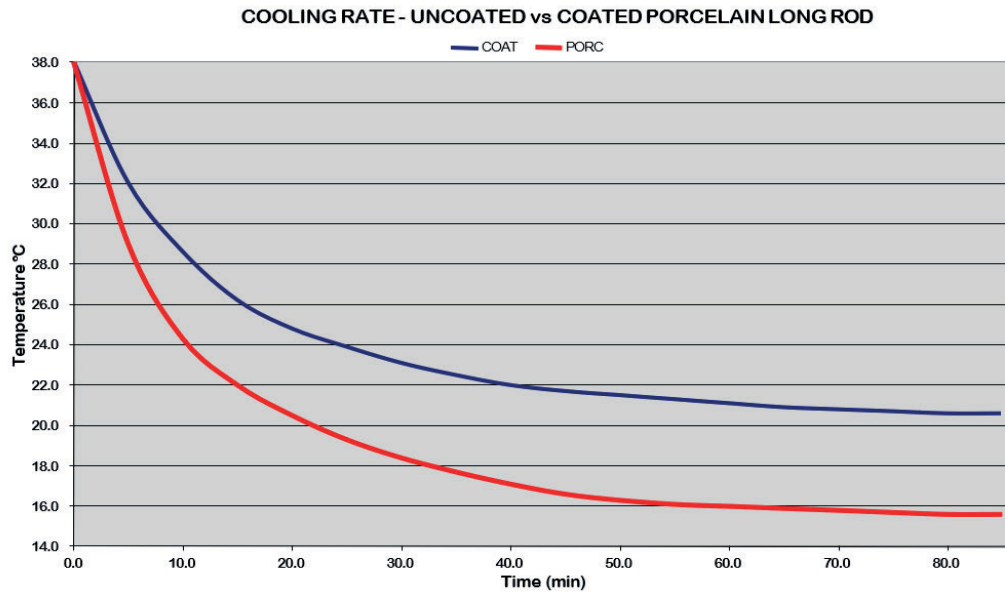
## BONDING AGENT CONTENT

Silicone rubber does not naturally adhere to other materials – hence its widespread use in the manufacture of moulds. Thus, in most coatings, a bonding agent is included. This means that the percentage of silicone in the coating is reduced and a significant volume of bonding agent is dispersed within the coating layer where it serves no beneficial purpose. To further increase the silicone percentage, with the CharCoat IC system a primer is applied separately prior to coating application, therefore the bonding agent exists only where it is required, i.e. at the interface between the coating and the porcelain, and no extraneous material is displacing the silicone content.

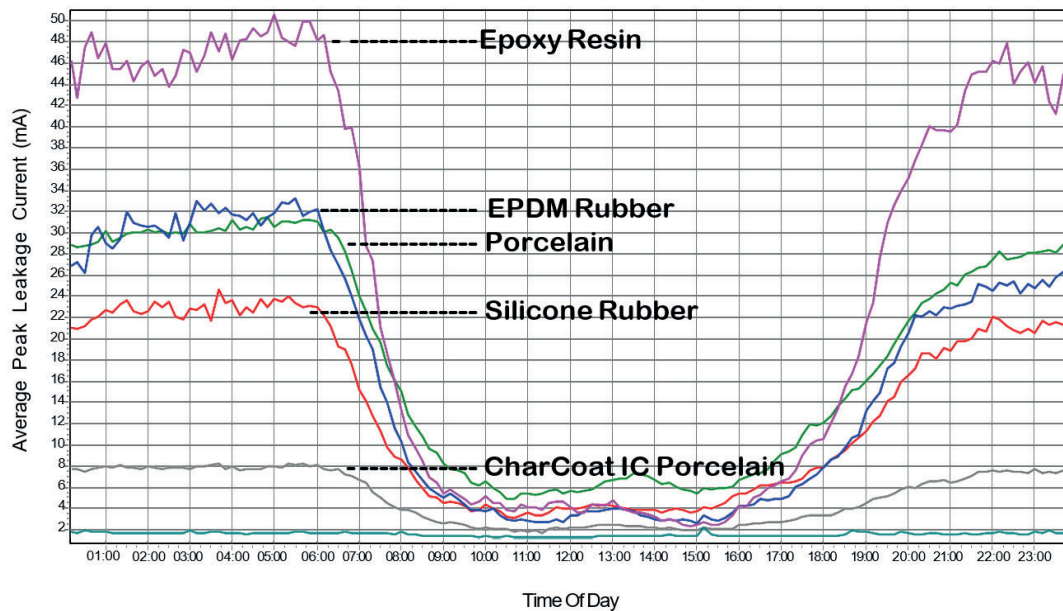
## THERMAL CHARACTERISTICS

As stated in the IEEE Standard, to prevent premature aging of the coating, it is important that the surface heating created by electrical discharges is efficiently conducted away. As silicone rubber has a high thermal resistance, the coating thickness has an important influence on this. The diagrams below show the insulator temperature change when coated and uncoated porcelain insulators are externally heated and cooled. This illustrates the high thermal resistance of silicone rubber and why thermal effects need to be considered.





In addition to life extension, it has been found that thinner coatings also offer superior performance to thicker ones. This was best illustrated by a PhD thesis undertaken by Dr. W L Vosloo of Eskom. His studies focused on the behavior of insulators of identical shape and differing only in the material of their construction. A comparison of the average leakage current amplitudes recorded over a one-year period in the severe, natural environment of the Koeberg Test Station is depicted below.



As can be noted from the diagram, the porcelain insulator coated with CharCoat IC displayed the lowest leakage current activity. More surprising, however, is the fact that the daily maximum current amplitude on the CharCoat IC porcelain was less than 35% of that of the silicon rubber composite unit. This large disparity is not due to their surface properties – both being fully hydrophobic – but due to their different thermal characteristics. The thin coating layer on a porcelain substrate allowed the temperature of the coated insulator to more closely track the changes in the ambient temperature. The composite unit with no benefit from a porcelain heat sink experienced greater condensation and consequent wetting of the surface contamination, resulting in increased leakage current flow.

## **CHARCOAT IC FORMULATION**

In view of the theoretical considerations and practical test results described above, CharCoat IC was formulated with three different solvent types to facilitate the easy achievement of a thin coating layer with a smooth surface to inhibit contaminant accumulation. In addition, silicone content was maximized by optimizing the silicone: filler ratio and removing the bonding agent from the material.

## **APPLICATION PHILOSOPHY**

Based on the valuable research conducted at Koeberg and subsequent experience from many sites in the field, the recommended average thickness of an CharCoat IC coating layer to realise adequate heat dissipation, was defined as 0.2mm after complete curing. This translates to a coating usage of 1 litre per 2.3m<sup>2</sup> of insulator surface.

Because of the fundamental geometry of the insulator, in practice there will always be variation of the layer thickness over the insulator surface. For example, in spraying the small valley between the ridges on the shed undersides, considerably more coating will build up on the edges of the ridges and sheds. However, such layer non-uniformity has negligible influence on the insulator performance. The coating is undertaken for the sole purpose of giving the insulator hydrophobic properties. These are not dependent on the coating thickness. Secondly, heat dissipation relates more to the average thickness than that at isolated points. It must also be appreciated that, if a minimum thickness is specified, a major part of the insulator surface will have a thickness far above this value resulting in wider thickness variation and much material wastage.

Measuring dry coating thickness – particularly over many points to obtain a minimum or average – is difficult. Ultrasonic measurements are not very accurate and physical methods are destructive, necessitating the need for subsequent touch-ups. Measuring thickness with the coating in a wet state is easier but not precise in defining the ultimate dry thickness as the amount of solvent evaporation prior to the measurement is unknown. Moreover, the degree of evaporation is highly dependent on the ambient temperature so the measured wet to dry thickness ratio will vary with the ambient conditions.

For these reasons, it is recommended that application control be exercised by measuring the total amount of material used per unit area of insulator surface.

## **TEST AND SERVICE EXPERIENCE**

The formulation of CharCoat IC and its application method has proved to be successful by the excellent performance experienced over more than 25 years on insulators ranging from 11kV to 765kVAC and 25kv to 533kVDC in the most severe marine, industrial, desert and agricultural environments in Asia, Africa, Australasia, Europe, North America and the Middle East.

To quantify the performance of CharCoat IC in comparison to other coatings, reference can be made to the diagram below. The graph indicates the accumulated leakage current activity recorded at the Koeberg Test Station over a one year period on three identical porcelain insulators – two of which were coated with different coatings and one left uncoated. (Details of the test procedure and assessment method are defined in the Eskom Standard 34-224). In order for a product to pass the test and gain approval for use on the Eskom system, no more than three flashovers are permitted over the one year test period.

As shown in the Figure, the uncoated porcelain had the highest leakage current activity and experienced its fourth flashover after 6 months. Coating Type 2, a widely-used, thick, high filler content material complying fully with the MEW, Kuwait and SEC, Saudi Arabia Specifications – failed after suffering its fourth flashover in the ninth month. CharCoat IC displayed far lower leakage current amplitudes and did not have a single flashover during the full year test period. CharCoat IC is the only coating to have achieved this distinction to date.

